Annotated Bibliography

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[1] Thorsten Beckmann and Pieter Belmans. Homological projective duality for the segre cubic. (arXiv:2202.08601), February 2022. arXiv:2202.08601 [math]

- Very well explained example of HPD, one step up in complexity from intersection of quadrics vs hyperelliptic curve, which is duality between the (small resolution of) segre cubic S_3 and (small resolution of) double cover $\pi : \mathrm{Cob} \to \mathbb{P}^4$ branched along Castelnuovo-Richmond quartic CR_4.
- The S_3 has interesting linear sections, including cubic surfaces, Kummer quartics, elliptic curves. The linear sections of the dual also admit classical descriptions.
- Data for HPD is a map $X \to \mathbb{P}V$ *and* a Lefschetz decomposition. The HPD dual described above is for a certain rectangular Lefschetz decomposition (which is maximal in some sense, very good), but there is also another decomposition coming from a different small resolution of S_3, coming from a quiver moduli description of S_3 as in [10]. Specifically, S_3 is a moduli space of quiver representations for a stability condition lying on a wall, and small perturbations give small desingularizations, which (through some sort of mutations?) give new Lefschetz decompositions.
- **Further work:** understand (noncommutative) HPD for this new Lefschetz center.

[2] Pieter Belmans, Jishnu Bose, Sarah Frei, Benjamin Gould, James Hotchkiss, Alicia Lamarche, Jack Petok, Cristian Rodriguez Avila, and Saket Shah. On decompositions for fano schemes of intersections of two quadrics

- Proposes conjectural semiorthogonal decompositions for Fano schemes of linear subspaces on intersections of two quadrics (denote $F_k(Q_1 \cap Q_2)$, where $Q_1 \cap Q_2$ is in $\mathbb{P}^{2g+1}$ or $\mathbb{P}^{2g}$), in terms of symmetric powers of an associated hyperelliptic/stacky curve (for odd/even cases). Main evidence is a conjectured identity in the Grothendieck ring of varieties, which in turn is suggested by a (conjectured) equality of E-polynomials (assembles Hodge numbers but makes sense in non smooth/proper setup) which is achieved by upgrading recent topological results from [4, 5] to Hodge structures. Upgrading the result involves going back through the proof (Springer theory, mixed Hodge modules, perverse sheaves).
- This is a somewhat reasonable conjecture to make (i.e., it implies the evidence) because

of the following commutative diagram:

$$\begin{array}{ccccc} K_0(\mathrm{Var}/k) & \xrightarrow{\mu_{\mathrm{Hdg}}} & K_0(\mathrm{HS}) & \xrightarrow{\mu_{\mathrm{E}}} & \mathbb{Z}[x,y] \\ \downarrow{\scriptstyle \mu_{D^b}} & & & & \downarrow \\ K_0(\mathrm{Cat}/k) & & \xrightarrow{\mu_{\mathrm{Hoch}}} & & \mathbb{Z}[t,t^{-1}] \end{array}$$

- The ring $K_0(\mathrm{Cat}/k)$ is the *Grothendieck ring of categories*, i.e., a quotient of the free abelian group of smooth and proper pretriangulated dg categories, where you mod out by the "cut-and-paste-relation", setting $[\mathcal{A}] = [\mathcal{B}] + [\mathcal{C}]$ whenever you have a semiorthogonal decomposition of the corresponding triangulated categories $H^0(\mathcal{A}) = \langle H^0(\mathcal{B}), H^0(\mathcal{C})\rangle$. You get a ring structure by tensor product of dg categories.
- The ring $K_0(HS)$ is the Grothendieck ring of the category of polarizable pure rational Hodge structures.
- μ_{Hdg} sends a variety to its Hodge structure.
- μ_E sends a Hodge structure to its Hodge polynomial $\sum_{p,q=0}(-1)^{p+q}h^{p,q}x^py^q \in \mathbb{Z}[x,y]$
- μ_{D^b} sends a smooth proper variety to $D^b(X)$ (and extended in a reasonable way by "Bittner's presentation" and "Orlov's blowup formula").
- μ_{Hoch} sends a dg category to its Hochschild homology polynomial $\sum \dim_k HH_i t^i$ (can have negative degrees).
- The Hochschild-Kostant-Rosenberg decomposition

  $$HH_i(X) \cong \bigoplus_{q-p=i} H^p(X, \Omega_X^q)$$

  implies the diagram commutes, where we map (I think??)

  $$\begin{aligned} \mathbb{Z}[x,y] &\longmapsto \mathbb{Z}[t,t^{-1}] \\ q &\longmapsto t \\ p &\longmapsto t^{-1} \end{aligned}$$

- This conjecture is known in a few cases:
  - If $k = 0$, the conjecture is known by Bondal-Orlov/Kuznetsov.
  - If $k = 0, \ldots, g-2$, embeddings of $D^b(C)$ into $D^b(F_k(Q_1 \cap Q_2))$ are known by [9][Theorem 2.11].
  - If $k = g-2$ and odd dimensional projective space, this is known by the (now proven) BGMN conjecture [14, 13] (this is a crazy paper!), due to moduli description by [8].

[3] Pieter Belmans and Andreas Krug. Derived categories of (nested) hilbert schemes. *Michigan Mathematical Journal*, 74(1), February 2024. arXiv:1909.04321 [math]

- Let S be a smooth projective surface, and let $\mathcal{I}$ be the ideal sheaf of the universal subscheme on $S \times S^{[n]}$. Denote F the Fourier-Mukai functor $\Phi_{\mathcal{I}} : D^b(S) \to D^b(S^{[n]})$. Then, they prove for $n \geq 2$
  - F is fully faithful iff $H^1(S, \mathcal{O}_S) = H^2(S, \mathcal{O}_S) = 0$ (i.e., $\mathcal{O}_S$ is exceptional) (this is a reverse implication of a known result).
  - F is a $\mathbb{P}^m$-functor or a spherical functor iff S is a K3 surface, where define a $\mathbb{P}^n$-functor
  - If $D^b(S)$ has an exceptional collection using line bundles L_i, $D^b(S^{[n]})$ gets a semiorthogonal decomposition using associated line bundles on $S^{[n]}$.
  - They get a simplified proof of Toda's semiorthogonal decomposition of a symmetric power of a curve using Jacobians.
  - They get a semiorthogonal decomposition
  $$D^b(S^{[n-1,n]}) = \langle D^b(S \times S^{[n-1]}, D^b(S^{[n-2,n-1]})) \rangle$$
  which one can expand inductively. No assumptions on S needed!
- They use derived McKay correspondence, G-equivariant derived category, quotient stack.
- Something still works in higher dimensions if you use a symmetric quotient stack instead of the Hilbert scheme.
- Spherical means cone over unit is auto-equivalence which composes to be right adjoint... don't have any intuiting.
- A $\mathbb{P}^n$-functor is similar, but has to do with a $\mathbb{P}$-cotwist auto-equivalence H.
- Use Jiang-Leung's "generalized blowup formula" [11].

[6] Raymond Cheng, Alexander Perry, and Xiaolei Zhao. Derived categories of quartic double fivefolds. (arXiv:2403.13463), April 2025. arXiv:2403.13463 [math]

- Kuznetsov's conjecture that a cubic fourfold $X \subseteq \mathbb{P}^5$ is rational iff $\mathcal{K}u(X)$ is a K3 surface comes from heurstics that suggest the following more general claim: if $\mathcal{K}u(X)$ is a CY category of dimension $\dim X - 2$ and X is rational, then $\mathcal{K}u(X)$ is equivalent to the D^b of a smooth projective CY variety.
- This paper investigates an interesting 5-dimensional example, taking a double cover $X \to \mathbb{P}^5$ branched along a quartic hypersurface that singular along a line $L \subseteq \mathbb{P}^5$. They prove that there is a (weak) crepant categorical resolution of singularities of $\mathcal{K}u(X)$ by $D^b(W^+, \mathcal{A}^+)$, where W^+ is a proper 3-dim CY algebraic space which is not projective, and $\mathcal{A}^+$ is an Azumaya algebra on W^+ whose Brauer class is nontrivial. See [7] for twisting by Brauer classes (which can be represented by Azumaya algebras), and [12] for the definition of a (weak) crepand categorical resolution.

- The proof studies a RoS $\widetilde{X} \to X$, and gets a Kuznetsov component $\widetilde{\mathcal{K}u}(X) \subseteq D^b(\widetilde{X})$ which is a crepant categorical resolution of $\mathcal{K}u(X)$, and uses the quadric bundle structure (look into this) on X induced by linear projection from L to identify $\widetilde{\mathcal{K}u}(X)$ as the derived category of $(W^+, \mathcal{A}^+)$.
- Another theorem when the quadric bundle admits sections and more singularities you get a projective CY 3-fold.

[12] Alexander Kuznetsov. Semiorthogonal decompositions in algebraic geometry. (arXiv:1404.3143), January 2015. arXiv:1404.3143 [math]

- A **categorical resolution of singularities** of a scheme Y is a "smooth" triangulated category $\mathcal{T}$ and an adjoint pair of triangulated functors $\pi_* : \mathcal{T} \to D^b(Y)$ and $\pi^* : D^{\mathrm{perf}}(Y) \to \mathcal{T}$ such that $\pi_* \circ \pi^* \cong \mathrm{id}_{D^{\mathrm{perf}}(Y)}$ (in particular, π^* is fully-faithful).
- A categorical resolution $(\mathcal{T}, \pi_*, \pi^*)$ of a scheme Y is **weakly crepant** if π^* is in addition a right adjoint to π_*. (generalizing $K_{\widetilde{Y}/Y} = 0$). This is the notion referred to in [6].
- A categorical resolution is **strongly crepant** if the relative Serre functor of $\mathcal{T}$ over $D^b(Y)$ is isomorphic to identity.
- MMP for categorical resolutions seems to be easier.

[14] Jenia Tevelev and Sebastián Torres. The bgmn conjecture via stable pairs. (arXiv:2108.11951), November 2023. arXiv:2108.11951 [math]

- Let C be a smooth complex projective curve of genus $g \geq 2$, and $N = M_C(2, \Lambda)$ the moduli space of stable vector bundles on C of rank 2 with fixed determinant Λ of odd degree. Then, N is a smooth Fano variety of index 2 and $\mathrm{Pic} N = \mathbb{Z} \cdots \theta$ for some ample line bundle θ. Then, the authors gives a semiorthogonal decomposition of $D^b(N)$ (prove semiorthogonality here, prove fullness in [13]).
- Decomposition is given by several copies of $D^b(\mathrm{Sym}^i C)$ for $i = 0, \ldots, g-2$ embedded by Fourier Mukai transforms with kernel given by the "Poincare bundles" $\mathcal{E}$ or $\overline{\mathcal{E}}$ on $C \times N$, then tensored by powers of θ^*.
- Here, by the Poincare bundles

  $$\mathcal{E} = F^{\boxtimes \alpha} \in D^b(\mathrm{Sym}^\alpha C \times M_i(\lambda)), \qquad \overline{\mathcal{E}} = \overline{F}^{\boxtimes \alpha} \in D^b(\mathrm{Sym}^\alpha C \times M_i(\Lambda))$$

  they mean the symmetric group invariant parts of box products/sgn-twisted box products of F, the universal bundle on $C \times M_i(\Lambda)$, where $M_i(\Lambda)$ is a moduli space of stable pairs (E, φ) (where E is a rank 2 vector bundle, $\varphi \in H^0(E)$ a nonzero section) with varying GIT stability conditions for different i. The derived categories of these $M_i(\Lambda)$ then embed into succesive ones across wall crossings, and then there is a final map from $M_{g-1} \to N$ is the "Abel-Jacobi" map with the fiber over a vector bundle given by $\mathbb{P}H^0(C, E)$.

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[14] Jenia Tevelev and Sebastián Torres. The bgmn conjecture via stable pairs. (arXiv:2108.11951), November 2023. arXiv:2108.11951 [math].